11.10 Orthogonal Polynomials

Section 2 ran Gram–Schmidt on $(1, x, x^2)$ with $\langle p, q\rangle = \int_{-1}^{1} pq$, and stopped at degree 2 only because the space did (Example (Gram–Schmidt on $1, x, x^2$)). Nothing in the process cared. Feed it $(1, x, x^2, x^3, \dots)$ and it returns an infinite orthogonal sequence, one polynomial of each degree. Three things make such sequences worth a section. Each member is computable from the two before it. Each member of degree k has k distinct real roots inside the interval. And those roots are the best possible points at which to sample a function to estimate its integral. The first fact rests on one small identity; the other two rest on the first.

Throughout this section the field is $\mathbb{R}$.

11.10.1 Two ways to make an inner product on polynomials

Both constructions we need appeared in Section 1.

From a weight. Fix an interval with endpoints $-\infty \le a < b \le \infty$ and a **weight** w: a function continuous and **strictly positive** on the open interval (a, b), with $\int_a^b |t|^k w(t)\,\mathrm{d}t$ finite for every $k \in \mathbb{N}$. Define

$$\langle p, q\rangle = \int_a^b p(t)\,q(t)\,w(t)\,\mathrm{d}t.$$

Finiteness of the moments makes this a real number for all $p, q \in \mathbb{R}[x]$, and bilinearity and symmetry come from linearity of the integral. For positivity, let $p \neq 0$. Then $p^2 w$ is continuous and ≥ 0 on (a, b), and > 0 off the finitely many roots of p (Corollary 6.1.4), so $\langle p, p\rangle > 0$. That last step is the one calculus fact this section leans on: a continuous nonnegative function on an interval whose integral is 0 vanishes identically. As in Definition 11.1.1 it follows from the local fact that a continuous function nonzero at a point stays bounded away from 0 on a small interval around it, which then contributes a positive amount to the integral.

From nodes. Fix distinct $c_0 < \cdots < c_n$ in $\mathbb{R}$ and put $\langle p, q\rangle = \sum_{i=0}^{n} p(c_i)q(c_i)$. Section 1 checked that this is an inner product **on** $\mathbb{R}[x]_{\le n}$, positivity coming from the root bound. It is not one on $\mathbb{R}[x]$: the polynomial $(x - c_0)\cdots(x - c_n)$ is nonzero and pairs with itself to give 0. So in the node case everything below stops at degree n.

The two constructions share one property, and it is the engine of the section.

Proposition 11.10.1. Multiplication by x is symmetric

Let $\langle\cdot,\cdot\rangle$ be an inner product on $\mathbb{R}[x]$ coming from a weight, or the node inner product on $\mathbb{R}[x]_{\le n}$. Then

$$\langle xp, q\rangle = \langle p, xq\rangle$$

for all polynomials p, q in the space (in the node case, for all p, q of degree at most $n - 1$, so that xp and xq still lie in it).

Proof. In the weight case both sides equal $\int_a^b t\,p(t)q(t)w(t)\,\mathrm{d}t$, and in the node case both sides equal $\sum_{i=0}^{n} c_i\,p(c_i)q(c_i)$.

■

The proof is one line, but the statement is not a triviality: multiplication by x moves from one slot to the other for free. That is exactly the identity $\langle Tu, v\rangle = \langle u, Tv\rangle$ defining a **self-adjoint** operator (Definition 11.6.2), and it is the reason the recurrence below has three terms instead of $k+2$.

Warning. **Multiplication by x is not an operator on $\mathbb{R}[x]_{\le n}$.** It raises degree, so it maps that space out of itself; on $\mathbb{R}[x]$ it is an operator, but there the space is infinite-dimensional, and Definition 11.6.2 was stated for a finite-dimensional one. "Self-adjoint" is here a name for Proposition 11.10.1, not a literal application of the definition. It is the identity, not the name, that the proofs use.

11.10.2 Orthogonal polynomial sequences

Applied to the independent list $(1, x, x^2, \dots)$, Gram–Schmidt never stalls. What comes out is one polynomial of each degree, pairwise orthogonal, and that combination deserves a name.

An orthogonal polynomial sequence is one polynomial of every degree, no two of them pairing to anything but zero.

Definition 11.10.1. Orthogonal Polynomial Sequence

Let $\langle\cdot,\cdot\rangle$ be an inner product on $\mathbb{R}[x]$, or on $\mathbb{R}[x]_{\le n}$. An **orthogonal polynomial sequence** for it is a sequence $p_0, p_1, p_2, \dots$ of polynomials (a finite list $p_0, \dots, p_n$ in the second case) such that

(OP1) $\deg p_k = k$ for **every** index k;
(OP2) $\langle p_j, p_k\rangle = 0$ whenever $j \neq k$.

The sequence is **monic** if every p_k has leading coefficient 1.

In words. (OP1) says one polynomial of each degree: no gaps, no repeats, no accidental drop in degree. (OP2) says the members are pairwise orthogonal. Together they say more than either alone, because by (OP1) the list $(p_0, \dots, p_{k-1})$ is a basis of $\mathbb{R}[x]_{\le k-1}$, so (OP2) upgrades to

$$\langle p_k, q\rangle = 0 \qquad \text{for every } q \in \mathbb{R}[x]_{\le k-1},$$

and that is the form in which the condition always gets used.

Two things need checking: that such a sequence exists, and that fixing the scale pins it down.

Proposition 11.10.2. Existence and uniqueness of the monic sequence

Let $\langle\cdot,\cdot\rangle$ be an inner product on $\mathbb{R}[x]$. For each $k \in \mathbb{N}$ there is **exactly one** monic p_k of degree k with $\langle p_k, q\rangle = 0$ for every $q \in \mathbb{R}[x]_{\le k-1}$. These p_k form the unique monic orthogonal polynomial sequence, and $(p_0, \dots, p_k)$ is an orthogonal basis of $\mathbb{R}[x]_{\le k}$ for every k. The same holds on $\mathbb{R}[x]_{\le n}$ for $k \le n$.

Idea. Existence is Gram–Schmidt, once we know the k-th output really has degree k; it does, because it lies in $\mathbb{R}[x]_{\le k}$ but not in the span of its predecessors, which is $\mathbb{R}[x]_{\le k-1}$. Uniqueness is the standard move: two candidates differ by a polynomial of lower degree that

is orthogonal to everything of lower degree, hence to itself.

Proof. Existence. The list $(1, x, \dots, x^k)$ is independent, so Theorem 11.2.2 applies and returns an orthonormal list $(e_0, \dots, e_k)$ with $\operatorname{span}(e_0, \dots, e_j) = \operatorname{span}(1, x, \dots, x^j) = \mathbb{R}[x]_{\le j}$ for every $j \le k$. Fix j. Then $e_j \in \mathbb{R}[x]_{\le j}$, and $e_j \notin \mathbb{R}[x]_{\le j-1} = \operatorname{span}(e_0, \dots, e_{j-1})$, since an orthonormal list is independent (Theorem 11.1.4). Hence $\deg e_j = j$ exactly. Let $a_j \neq 0$ be its leading coefficient and put $p_j = e_j/a_j$, monic of degree j. For $q \in \mathbb{R}[x]_{\le j-1} = \operatorname{span}(e_0, \dots, e_{j-1})$ we get $\langle p_j, q\rangle = 0$ by orthogonality of the list.

Uniqueness. Suppose p and p' are both monic of degree k and orthogonal to every element of $\mathbb{R}[x]_{\le k-1}$. Their leading terms cancel, so $p - p' \in \mathbb{R}[x]_{\le k-1}$. Therefore $\langle p - p', p - p'\rangle = \langle p, p - p'\rangle - \langle p', p - p'\rangle = 0$, and positive definiteness gives $p = p'$.

Finally, $(p_0, \dots, p_k)$ is an orthogonal list of nonzero vectors, hence independent (Theorem 11.1.4), with $k + 1 = \dim \mathbb{R}[x]_{\le k}$ members, hence a basis (Theorem 1.6.4). ∎

Examples. For $\langle p, q\rangle = \int_{-1}^{1} pq$ the first three monic members are 1, x, $x^2 - \frac{1}{3}$ (Example (Gram–Schmidt on $1, x, x^2$)); these are the monic **Legendre polynomials**. For the node inner product with nodes $-1, 0, 1$ on $\mathbb{R}[x]_{\le 2}$ we have $\langle 1, 1\rangle = 3$, $\langle x, 1\rangle = 0$, $\langle x, x\rangle = 2$, $\langle x^2, 1\rangle = 2$, $\langle x^2, x\rangle = 0$, so $p_0 = 1$, $p_1 = x$, $p_2 = x^2 - \frac{2}{3}$ — and there the sequence ends, since $\mathbb{R}[x]_{\le 2}$ holds nothing of degree 3. The degenerate member is always $p_0 = 1$, the only monic polynomial of degree 0, with an empty condition on it.

A nonexample by minimal change. Replace the Legendre $p_2 = x^2 - \frac{1}{3}$ by $\tilde{p}_2 = x^2 + x - \frac{1}{3}$. Still monic of degree 2, so (OP1) survives; but $\langle \tilde{p}_2, x\rangle = \langle x, x\rangle = \frac{2}{3} \neq 0$, so (OP2) fails. Adding a multiple of a lower member is exactly the freedom orthogonality removes.

Why monic? Rescaling each p_k by a nonzero λ_k gives another orthogonal polynomial sequence, so a normalization has to be chosen, and traditions differ: the classical Legendre polynomials P_k are fixed by $P_k(1) = 1$, and Gram–Schmidt as run in Section 2 fixes $\|p_k\| = 1$. Monic is the convenient choice here, because it is the one for which the recurrence below needs no extra factor.

Warning. **The sequence belongs to the inner product, not to the polynomials.** "The orthogonal polynomials" is meaningless without a weight and an interval. Exercise B2 of Section 2 (Exercise (B2: Gram–Schmidt on polynomials)) ran the same process with $\int_0^1$ instead of $\int_{-1}^1$ and got 1, $x - \frac{1}{2}$, $x^2 - x + \frac{1}{6}$: same degrees, different polynomials.

Quick check. For $\langle p, q\rangle = \int_{-1}^{1} pq$, how many polynomials of degree 2 with $\|p\| = 1$ are orthogonal to $\mathbb{R}[x]_{\le 1}$? Write them down.

Solution. Exactly two. If p has degree 2 and $p \perp \mathbb{R}[x]_{\le 1}$, let $\lambda \neq 0$ be its leading coefficient; then p/λ is monic of degree 2 and still orthogonal to $\mathbb{R}[x]_{\le 1}$, so $p/\lambda = x^2 - \frac{1}{3}$ by Proposition 11.10.2. Thus $p = \lambda(x^2 - \frac{1}{3})$, and $\|p\| = 1$ leaves $\lambda = \pm 1/\|x^2 - \frac{1}{3}\|$. Since Example

(Gram–Schmidt on $1, x, x^2$) computed $\|x^2 - \frac{1}{3}\|^2 = \frac{8}{45}$, the two answers are $\pm\frac{\sqrt{10}}{4}(3x^2 - 1)$. Fixing the scale by the norm leaves a sign; fixing it by "monic" leaves nothing.

11.10.3 The three-term recurrence

Computing p_n by Gram–Schmidt costs one inner product for every earlier member, so reaching degree n costs about $n^2/2$ integrals. Almost all of them return 0. The next theorem says which ones do not, and reduces the cost to two integrals per step.

Theorem 11.10.3. Three-Term Recurrence

Let (p_k) be the monic orthogonal polynomial sequence of Proposition 11.10.2, for an inner product satisfying Proposition 11.10.1. For a weight on an interval this covers every k; for the node inner product on n nodes, where Proposition 11.10.1 holds only up to degree $n-1$, read the statement for $k+1 \le n$. Put $p_{-1} = 0$ and, for $k \ge 0$,

$$a_k = \frac{\langle xp_k, p_k\rangle}{\langle p_k, p_k\rangle}, \qquad b_k = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle} \quad (k \ge 1).$$

Then $p_0 = 1$, $p_1 = x - a_0$, and

$$p_{k+1} = (x - a_k)\, p_k - b_k\, p_{k-1} \qquad \text{for every } k \ge 1.$$

Moreover $b_k > 0$ for every $k \ge 1$.

Idea. The polynomial xp_k is monic of degree $k+1$, so it is p_{k+1} plus something of degree at most k, and that something expands in the orthogonal basis $(p_0, \dots, p_k)$ with coefficients $\langle xp_k, p_j\rangle / \langle p_j, p_j\rangle$ (Corollary 11.3.4). The point is that almost every coefficient is 0: move the x across with Proposition 11.10.1, and the coefficient becomes $\langle p_k, xp_j\rangle$, which vanishes as soon as $\deg(xp_j) = j+1$ is still below k. Only $j = k$ and $j = k-1$ can survive.

Proof. $p_0 = 1$ is the unique monic polynomial of degree 0, and $p_1 = x - a_0$ is the unique monic polynomial of degree 1 with $\langle p_1, p_0\rangle = 0$, since $\langle x - a_0, 1\rangle = \langle x, 1\rangle - a_0\langle 1, 1\rangle = 0$ exactly for $a_0 = \langle x \cdot 1, 1\rangle / \langle 1, 1\rangle$.

Now fix $k \ge 1$. By Proposition 11.10.2, $(p_0, \dots, p_{k+1})$ is an orthogonal basis of $\mathbb{R}[x]_{\le k+1}$, and xp_k lies in that space. By Corollary 11.3.4 applied inside $\mathbb{R}[x]_{\le k+1}$,

$$xp_k = \sum_{j=0}^{k+1} c_j\, p_j, \qquad c_j = \frac{\langle xp_k, p_j\rangle}{\langle p_j, p_j\rangle}.$$

Both xp_k and p_{k+1} are monic of degree $k+1$, and every other term of the sum has degree at most k; comparing coefficients of x^{k+1} gives $c_{k+1} = 1$.

Let $j \le k-2$. By Proposition 11.10.1, $\langle xp_k, p_j\rangle = \langle p_k, xp_j\rangle$, and $\deg(xp_j) = j+1 \le k-1$, so $xp_j \in \mathbb{R}[x]_{\le k-1}$ and the pairing is 0 by Proposition 11.10.2. Hence $c_j = 0$ for every $j \le k-2$, and

$$xp_k = p_{k+1} + c_k\, p_k + c_{k-1}\, p_{k-1},$$

which rearranges to $p_{k+1} = (x - c_k)p_k - c_{k-1}p_{k-1}$.

It remains to identify the two coefficients. By definition $c_k = \langle xp_k, p_k\rangle / \langle p_k, p_k\rangle = a_k$. For the other, xp_{k-1} is monic of degree k, so $xp_{k-1} - p_k$ has degree at most $k-1$ and therefore pairs to 0 with p_k. Hence, using Proposition 11.10.1 again,

$$\langle xp_k, p_{k-1}\rangle = \langle p_k, xp_{k-1}\rangle = \langle p_k, p_k\rangle,$$

so $c_{k-1} = \langle p_k, p_k\rangle / \langle p_{k-1}, p_{k-1}\rangle = b_k$. Finally b_k is a quotient of $\|p_k\|^2$ by $\|p_{k-1}\|^2$, both strictly positive because p_k and p_{k-1} are nonzero, so $b_k > 0$.

■

Two inner products per step generate the whole sequence. The positivity of b_k is not decoration either: it is the sign that drives the interlacing argument of Exercise (C1: Interlacing roots).

Remark. If the interval is symmetric, $(a, b) = (-c, c)$, and the weight is even, $w(-t) = w(t)$, then $a_k = 0$ for every k and $p_k(-x) = (-1)^k p_k(x)$. Induct on k: $p_0 = 1$ is even, and $a_0 = \langle x, 1\rangle / \langle 1, 1\rangle = 0$ because $t\,w(t)$ is odd, so $p_1 = x$ is odd. If p_{k-1} and p_k have parities $(-1)^{k-1}$ and $(-1)^k$, then $t\,p_k(t)^2 w(t)$ is odd, so $a_k = 0$, and $p_{k+1} = xp_k - b_k p_{k-1}$ is a combination of two polynomials of parity $(-1)^{k+1}$.

Example. The monic Legendre polynomials, continued

For $\langle p, q\rangle = \int_{-1}^{1} pq$, Example (Gram–Schmidt on $1, x, x^2$) gave $p_0 = 1$, $p_1 = x$, $p_2 = x^2 - \frac{1}{3}$, with $\langle p_0, p_0\rangle = 2$, $\langle p_1, p_1\rangle = \frac{2}{3}$, and $\langle p_2, p_2\rangle = \frac{8}{45}$. Use Theorem 11.10.3 to obtain p_3, and check the recurrence reproduces p_2.

Solution. The weight $w = 1$ is even on $(-1, 1)$, so every $a_k = 0$ by the remark above. Then

$$b_1 = \frac{\langle p_1, p_1\rangle}{\langle p_0, p_0\rangle} = \frac{2/3}{2} = \frac{1}{3}, \qquad b_2 = \frac{\langle p_2, p_2\rangle}{\langle p_1, p_1\rangle} = \frac{8/45}{2/3} = \frac{4}{15}.$$

The recurrence at $k = 1$ gives $p_2 = x \cdot x - \frac{1}{3} \cdot 1 = x^2 - \frac{1}{3}$, which is what Gram–Schmidt produced. At $k = 2$,

$$p_3 = x\left(x^2 - \tfrac{1}{3}\right) - \tfrac{4}{15}\, x = x^3 - \tfrac{3}{5}x.$$

Check. $\langle p_3, 1\rangle = \langle p_3, x^2\rangle = 0$ because the integrands are odd, and

$$\langle p_3, x\rangle = \int_{-1}^{1} \left(t^4 - \tfrac{3}{5}t^2\right) \mathrm{d}t = \tfrac{2}{5} - \tfrac{3}{5} \cdot \tfrac{2}{3} = 0.$$

So $p_3 \perp \mathbb{R}[x]_{\le 2}$, as it should be. Two integrals produced a polynomial that Gram–Schmidt would have charged three for, and the saving grows with the degree.

11.10.4 Where the roots are

The roots of $p_2 = x^2 - \frac{1}{3}$ are $\pm 1/\sqrt{3} \approx \pm 0.577$, and those of $p_3 = x^3 - \frac{3}{5}x$ are 0 and $\pm\sqrt{3/5} \approx \pm 0.775$. All five lie inside $(-1, 1)$, where the weight lives. No coincidence. The argument first needs one fact about real polynomials, in which $\mathrm{mult}_r(f)$ is the multiplicity of r as a root of f

(Definition 6.4.1).

Lemma 11.10.4. Even multiplicities mean no sign change

Let $I \subseteq \mathbb{R}$ be an open interval and let $f \in \mathbb{R}[x]$ be nonzero. If $\mathrm{mult}_r(f)$ is even for every $r \in I$, then $f \geq 0$ throughout I, or $f \leq 0$ throughout I.

Proof. By Theorem 6.4.3 over $\mathbb{R}$, write $f = (x-d_1)^{m_1} \cdots (x-d_l)^{m_l}\, g$, where $d_1, \ldots, d_l$ are the distinct real roots of f, $m_i = \mathrm{mult}_{d_i}(f)$, and g has no real root. Separate the factors according to whether $d_i \in I$:

$$f = h \cdot \Big(\textstyle\prod_{d_i \in I} (x-d_i)^{m_i}\Big), \qquad h = g \cdot \textstyle\prod_{d_i \notin I} (x-d_i)^{m_i}.$$

Each m_i with $d_i \in I$ is even by hypothesis, so the second factor is the square of a real polynomial and is ≥ 0 everywhere. The polynomial h has no root in I: its roots are the d_i outside I, and g has none. Being continuous and nowhere zero on the interval I, h has constant sign there by the intermediate value theorem. Hence f has that same sign throughout I, weakly, with equality only at the d_i inside I.

■

Theorem 11.10.5. Roots of an Orthogonal Polynomial

Let w be a weight on (a,b), let $\langle p, q\rangle = \int_a^b pqw$, and let (p_k) be the monic orthogonal polynomial sequence. For every $k \geq 1$, the polynomial p_k has k **distinct real** roots, and every one of them lies in the **open** interval (a,b).

Idea. Suppose p_k changed sign at only a few points $c_1, \ldots, c_m$ of (a,b), with $m < k$. Build the polynomial $q = (x-c_1)\cdots(x-c_m)$. It is cheap — degree at most $k-1$ — so p_k is orthogonal to it. But q was designed to change sign at exactly the points where p_k does, so the product $p_k q$ changes sign nowhere, and a function of one sign that is not identically zero cannot integrate to zero against a positive weight. The contradiction forces $m \geq k$, and k distinct roots is all a polynomial of degree k can hold.

Proof. Fix $k \geq 1$. Let $c_1 < \cdots < c_m$ be the roots of p_k in (a,b) whose multiplicity is **odd** (possibly $m = 0$), and set $q = (x-c_1)\cdots(x-c_m)$, with $q = 1$ when $m = 0$.

Suppose, for contradiction, that $m \leq k-1$. Then $q \in \mathbb{R}[x]_{\leq k-1}$, so $\langle p_k, q\rangle = 0$ by Proposition 11.10.2.

Put $f = p_k q$, a nonzero polynomial (Theorem 0.8.3). Let $r \in (a,b)$ be a root of f. By Theorem 6.4.2, $\mathrm{mult}_r(f) = \mathrm{mult}_r(p_k) + \mathrm{mult}_r(q)$. If $r = c_i$ for some i, then $\mathrm{mult}_r(q) = 1$ and $\mathrm{mult}_r(p_k)$ is odd by the choice of the c_i, so the sum is even. If r is none of the c_i, then $q(r) \neq 0$, so $\mathrm{mult}_r(q) = 0$ and $\mathrm{mult}_r(p_k)$ is even, again by the choice of the c_i. Either way $\mathrm{mult}_r(f)$ is even.

By Lemma 11.10.4, $f \geq 0$ throughout (a,b) or $f \leq 0$ throughout (a,b); replacing q by $-q$ if necessary, which changes neither $\langle p_k, q\rangle = 0$ nor $\deg q$, we may assume $f \geq 0$ there. Then fw

is continuous and ≥ 0 on (a, b), and it is > 0 off the finitely many roots of f, because $w > 0$. By the calculus fact recalled when the inner product was set up — a continuous nonnegative function with integral 0 vanishes identically — an integrand of that shape cannot integrate to 0. Therefore

$$\langle p_k, q \rangle = \int_a^b f(t)\, w(t)\, \mathrm{d}t > 0,$$

contradicting $\langle p_k, q \rangle = 0$.

Hence $m \geq k$. Each c_i is a root of p_k, and p_k is nonzero of degree k, so $m \leq k$ by Corollary 6.1.4; thus $m = k$ and p_k has k distinct roots $c_1, \ldots, c_k$, all in (a, b). By Corollary 6.1.4 again, $(x - c_1) \cdots (x - c_k)$ divides p_k, and both are monic of degree k, so $p_k = (x - c_1) \cdots (x - c_k)$. In particular each root is simple and p_k has no roots outside (a, b), real or complex. This proves the theorem.

■

So the weight hands us k points inside its interval, chosen by the inner product rather than by us. The next subsection says what they are good for.

Quick check. The monic Laguerre polynomial of degree 2 for the weight $w(t) = e^{-t}$ on $(0, \infty)$ is $p_2 = x^2 - 4x + 2$. Check that Theorem 11.10.5 is not lying.

Solution. The roots are $2 \pm \sqrt{2}$, that is ≈ 0.586 and ≈ 3.414: real, distinct, and both in $(0, \infty)$. An unbounded interval is no obstacle, since the theorem asks only that the weight be positive with finite moments, and $\int_0^\infty t^k e^{-t}\, \mathrm{d}t = k!$.

11.10.5 Gaussian quadrature

Chapter 6 wrote $\int_0^1 p$ as a fixed combination of $p(0)$, $p(\frac{1}{2})$ and $p(1)$, valid for every $p \in \mathbb{R}[x]_{\leq 2}$ (Example (Integrals and Slopes from Values)). The construction works for any prescribed nodes: integrate the Lagrange interpolation formula (Theorem 6.6.1) and k distinct nodes give a rule exact on $\mathbb{R}[x]_{\leq k-1}$.

But a k-point rule has $2k$ numbers at its disposal, k nodes and k coefficients, and only k of them were spent. Spending the nodes too should buy degree $2k - 1$. Where to put them is the question, and the answer is the one place we have not chosen: the roots of p_k.

Theorem 11.10.6. Gaussian Quadrature

Let w be a weight on (a, b) with $\langle p, q \rangle = \int_a^b pqw$, let $k \geq 1$, and let $t_1 < \cdots < t_k$ be the roots of the monic orthogonal polynomial p_k, which lie in (a, b) by Theorem 11.10.5. Let $\ell_1, \ldots, \ell_k \in \mathbb{R}[x]_{\leq k-1}$ be the Lagrange polynomials for these nodes (Definition 6.6.1) and set

$$\alpha_i = \int_a^b \ell_i(t)\, w(t)\, \mathrm{d}t \qquad (i = 1, \ldots, k).$$

Then every α_i is **strictly positive**, and

$$\int_a^b f(t)\, w(t)\, \mathrm{d}t = \sum_{i=1}^{k} \alpha_i\, f(t_i)$$

for **every** $f \in \mathbb{R}[x]_{\leq 2k-1}$.

Idea. Two moves, one for each side of the identity. On the left, divide f by p_k: $f = qp_k + r$. The degree bookkeeping — $\deg f \leq 2k-1$ against $\deg p_k = k$ — forces $\deg q \leq k-1$, which is exactly the condition under which $\int qp_k w = \langle p_k, q\rangle$ vanishes. So the integral of f is the integral of its remainder. On the right, p_k vanishes at every node, so the sum also only sees the remainder. And the remainder has degree at most $k-1$, which is precisely the range on which k nodes reproduce a polynomial exactly. The two sides meet there.

Proof. Let $f \in \mathbb{R}[x]_{\leq 2k-1}$. Since $p_k \neq 0$, Theorem 6.1.1 gives $q, r \in \mathbb{R}[x]$ with

$$f = qp_k + r, \qquad \deg r < k.$$

If $q = 0$ then $q \in \mathbb{R}[x]_{\leq k-1}$ trivially. Otherwise $qp_k = f - r$ has degree at most $2k-1$, while $\deg(qp_k) = \deg q + k$ by Theorem 0.8.3; hence $\deg q \leq k-1$. Either way $q \in \mathbb{R}[x]_{\leq k-1}$, so Proposition 11.10.2 gives $\langle p_k, q\rangle = 0$, that is, $\int_a^b qp_k w = 0$. Therefore

$$\int_a^b fw = \int_a^b qp_k w + \int_a^b rw = \int_a^b rw. \tag{$*$}$$

At each node, $p_k(t_i) = 0$, so

$$f(t_i) = q(t_i)p_k(t_i) + r(t_i) = r(t_i) \qquad (i = 1, \ldots, k). \tag{$**$}$$

Since $\deg r \leq k-1$ and $t_1, \ldots, t_k$ are k distinct points, Theorem 6.6.1 (b) gives $r = \sum_i r(t_i)\ell_i$. Integrating against w and using the definition of α_i,

$$\int_a^b rw = \sum_{i=1}^{k} r(t_i)\, \alpha_i = \sum_{i=1}^{k} \alpha_i f(t_i),$$

the last step by $(**)$. Combining with $(*)$ proves the quadrature identity.

For positivity, fix j and apply the identity to $f = \ell_j^2$, which is legitimate because $\deg \ell_j^2 = 2k-2 \leq 2k-1$. By Theorem 6.6.1 (a), $\ell_j(t_i) = \delta_{ij}$, so the right-hand side collapses to α_j, while the left-hand side is $\langle \ell_j, \ell_j\rangle = \|\ell_j\|^2$. Since $\ell_j \neq 0$, positive definiteness gives $\alpha_j = \|\ell_j\|^2 > 0$.

■

Each hypothesis did one job. Orthogonality to lower degrees killed the quotient on the left; the nodes being roots of p_k killed it again on the right. That is also why the nodes cannot be moved: shift one and the second cancellation fails. Exercise (C2: The ceiling is 2k − 1) shows $2k-1$ is a ceiling, not just what this proof happens to reach.

Example. Two-point Gauss–Legendre

Take $w = 1$ on $(-1, 1)$ and $k = 2$. Find the nodes and weights, and test the resulting rule on $f = 1 + 2x + 3x^2 + 4x^3$ and on $f = x^4$.

Solution. The monic p_2 is $x^2 - \frac{1}{3}$ (Example (Gram–Schmidt on $1, x, x^2$)), so the nodes are $t_1 = -1/\sqrt{3}$ and $t_2 = 1/\sqrt{3}$. The Lagrange polynomials are $\ell_1 = (x - t_2)/(t_1 - t_2)$ and $\ell_2 = (x - t_1)/(t_2 - t_1)$, with $t_1 - t_2 = -2/\sqrt{3}$. Since $\int_{-1}^{1} t\,\mathrm{d}t = 0$ and $\int_{-1}^{1} 1\,\mathrm{d}t = 2$,

$$\alpha_1 = \frac{0 - 2t_2}{t_1 - t_2} = \frac{-2/\sqrt{3}}{-2/\sqrt{3}} = 1,$$

and $\alpha_2 = 1$ by the same computation with the roles swapped. The rule is

$$\int_{-1}^{1} f(t)\,\mathrm{d}t = f\Big(-\tfrac{1}{\sqrt{3}}\Big) + f\Big(\tfrac{1}{\sqrt{3}}\Big) \qquad \text{for } f \in \mathbb{R}[x]_{\le 3}.$$

Two evaluations, no coefficients to remember, every cubic exactly.

The cubic test. Exactly, $\int_{-1}^{1}(1 + 2t + 3t^2 + 4t^3)\mathrm{d}t = 2 + 0 + 3 \cdot \frac{2}{3} + 0 = 4$. By the rule, $f(-1/\sqrt{3}) = 2 - \frac{10\sqrt{3}}{9}$ and $f(1/\sqrt{3}) = 2 + \frac{10\sqrt{3}}{9}$, whose sum is 4. The odd parts cancel and the constant and quadratic parts contribute $1 + 1$ and $1 + 1$.

The quartic test. Exactly, $\int_{-1}^{1} t^4\,\mathrm{d}t = \frac{2}{5}$. By the rule, $(1/\sqrt{3})^4 + (1/\sqrt{3})^4 = \frac{2}{9}$. These differ, so degree $4 = 2k$ is already out of reach. Compare Example (Integrals and Slopes from Values), where the nodes were fixed in advance and **three** of them guaranteed exactness only up to degree 2 — guaranteed by the construction, that is; that particular symmetric rule happens to integrate cubics exactly as well, which is luck of the symmetry and not something the construction promises.

Warning. **"Exact" means exact on polynomials.** Theorem 11.10.6 says nothing about $\int_a^b fw$ for a general f, and nothing about what happens as k grows; on a nonpolynomial the rule returns an approximation whose error is a question for analysis. What linear algebra delivers is exactness on a subspace, and the reason that subspace is the largest available.

11.10.6 The classical families

Four weights account for most of the orthogonal polynomials that occur in practice. The table records them; the entries are statements, and Exercise (B2: Two rows of the table) asks you to verify two of the rows by integration.

Family	Interval	Weight $w(t)$	p_1	p_2
Legendre	$(-1, 1)$	1	x	$x^2 - \frac{1}{3}$
Chebyshev	$(-1, 1)$	$(1 - t^2)^{-1/2}$	x	$x^2 - \frac{1}{2}$
Hermite	$(-\infty, \infty)$	e^{-t^2}	x	$x^2 - \frac{1}{2}$
Laguerre	$(0, \infty)$	e^{-t}	$x - 1$	$x^2 - 4x + 2$

The last two columns give the monic members of degrees 1 and 2; $p_0 = 1$ in every row. The Chebyshev weight is unbounded at ± 1 and the Hermite and Laguerre intervals are unbounded, so in those three rows the integrals defining the inner product are improper; they converge, which is the finiteness condition our definition of a weight demanded. The first three weights are even on symmetric intervals, so their a_k all vanish and their members alternate in parity; the Laguerre weight is not, and $a_0 = 1$ there. Section 9 of this chapter meets the Chebyshev family again from the Fourier side, where the substitution $t = \cos\theta$ turns its awkward weight into $\mathrm{d}\theta$.

11.10.7 Exercises

11.10.7.1 A. Check your understanding

Exercise. A1

(a) Define what it means for $p_0, p_1, p_2, \dots$ to be a **monic orthogonal polynomial sequence** for an inner product on $\mathbb{R}[x]$.

(b) State the three-term recurrence, including the formulas for a_k and b_k and the starting values.

(c) True or false: $b_k > 0$ for every $k \geq 1$. Justify your answer.

(d) For $\langle p, q\rangle = \int_{-1}^{1} pq$, how many roots does p_7 have, of what kind, and where?

(e) True or false: for **every** inner product on $\mathbb{R}[x]$ one has $\langle xp, q\rangle = \langle p, xq\rangle$. Justify your answer.

(f) How many nodes does a Gaussian rule need in order to integrate every polynomial of degree at most 11 exactly?

Solution.

(a) A sequence with $\deg p_k = k$ and leading coefficient 1 for every k, and $\langle p_j, p_k\rangle = 0$ whenever $j \neq k$ (Definition 11.10.1).

(b) With $p_{-1} = 0$, $p_0 = 1$, and $p_1 = x - a_0$, one has $p_{k+1} = (x - a_k)p_k - b_k p_{k-1}$ for $k \geq 1$, where $a_k = \langle xp_k, p_k\rangle / \langle p_k, p_k\rangle$ and $b_k = \langle p_k, p_k\rangle / \langle p_{k-1}, p_{k-1}\rangle$ (Theorem 11.10.3).

(c) True. Both $\langle p_k, p_k\rangle = \|p_k\|^2$ and $\langle p_{k-1}, p_{k-1}\rangle = \|p_{k-1}\|^2$ are strictly positive, since p_k and p_{k-1} are nonzero and the inner product is positive definite.

(d) Seven roots, all real, all distinct, all in the open interval $(-1, 1)$ (Theorem 11.10.5). In particular p_7 has no complex roots and no repeated ones.

(e) False. Take the coefficient inner product $\langle p, q\rangle = \sum_k u_k v_k$ on $\mathbb{R}[x]$, where $p = \sum_k u_k x^k$ and $q = \sum_k v_k x^k$ (Example (A functional with no Riesz vector)). Then $\langle x \cdot 1, x\rangle = \langle x, x\rangle = 1$, while $\langle 1, x \cdot x\rangle = \langle 1, x^2\rangle = 0$. The identity is a property of the weight and node constructions (Proposition 11.10.1), not of inner products in general.

(f) Six. A k-point rule is exact up to degree $2k - 1$, and $2 \cdot 6 - 1 = 11$.

11.10.7.2 B. Practice

Exercise. B1: A weighted interval

On $\mathbb{R}[x]$ put $\langle p, q\rangle = \int_0^1 p(t)q(t)\,t\,\mathrm{d}t$. Compute the monic p_0, p_1, and p_2, and verify directly

that $p_2 \perp \mathbb{R}[x]_{\le 1}$.

Solution. The moments are $\langle x^k, 1\rangle = \int_0^1 t^{k+1}\mathrm{d}t = 1/(k+2)$, so $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ for $k = 0, 1, 2, 3$.

$p_0 = 1$. Then $a_0 = \langle x, 1\rangle / \langle 1, 1\rangle = (1/3)/(1/2) = \frac{2}{3}$, so $p_1 = x - \frac{2}{3}$, and

$$\langle p_1, p_1\rangle = \tfrac{1}{4} - \tfrac{4}{3}\cdot\tfrac{1}{3} + \tfrac{4}{9}\cdot\tfrac{1}{2} = \tfrac{1}{4} - \tfrac{2}{9} = \tfrac{1}{36}.$$

Next $\langle xp_1, p_1\rangle = \int_0^1 (t^4 - \frac{4}{3}t^3 + \frac{4}{9}t^2)\mathrm{d}t = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = \frac{2}{135}$, so

$$a_1 = \frac{2/135}{1/36} = \frac{8}{15}, \qquad b_1 = \frac{1/36}{1/2} = \frac{1}{18},$$

and Theorem 11.10.3 gives

$$p_2 = \left(x - \tfrac{8}{15}\right)\left(x - \tfrac{2}{3}\right) - \tfrac{1}{18} = x^2 - \tfrac{6}{5}x + \tfrac{3}{10}.$$

Verification. $\langle p_2, 1\rangle = \int_0^1 (t^3 - \frac{6}{5}t^2 + \frac{3}{10}t)\mathrm{d}t = \frac{1}{4} - \frac{2}{5} + \frac{3}{20} = 0$, and $\langle p_2, x\rangle = \int_0^1 (t^4 - \frac{6}{5}t^3 + \frac{3}{10}t^2)\mathrm{d}t = \frac{1}{5} - \frac{3}{10} + \frac{1}{10} = 0$. Since $(1, x)$ spans $\mathbb{R}[x]_{\le 1}$, this is the whole check.

Exercise. B2: Two rows of the table

Verify the degree-1 and degree-2 entries of the table for two of the classical families.

(a) Chebyshev: $w(t) = (1-t^2)^{-1/2}$ on $(-1, 1)$. You may use, without proof, that $\int_{-1}^1 (1 - t^2)^{-1/2}\mathrm{d}t = \pi$ and $\int_{-1}^1 t^2(1-t^2)^{-1/2}\mathrm{d}t = \pi/2$.

(b) Laguerre: $w(t) = e^{-t}$ on $(0, \infty)$. You may use, without proof, that $\int_0^\infty t^k e^{-t}\,\mathrm{d}t = k!$.

Solution.

(a) The weight is even, so $\langle x^j, x^i\rangle = 0$ whenever $i + j$ is odd. Hence $\langle x, 1\rangle = 0$, and x is monic of degree 1 and orthogonal to $\mathbb{R}[x]_{\le 0}$, so $p_1 = x$ by Proposition 11.10.2. Write $p_2 = x^2 + \beta x + \gamma$. Then $\langle p_2, x\rangle = \langle x^2, x\rangle + \beta\langle x, x\rangle + \gamma\langle 1, x\rangle = 0 + \beta\cdot\frac{\pi}{2} + 0$, forcing $\beta = 0$; and $\langle p_2, 1\rangle = \frac{\pi}{2} + 0 + \gamma\pi = 0$, forcing $\gamma = -\frac{1}{2}$. So $p_2 = x^2 - \frac{1}{2}$.

(b) The moments are $\langle x^k, 1\rangle = k!$, so $1, 1, 2, 6$ for $k = 0, 1, 2, 3$. For $p_1 = x + \beta$: $\langle p_1, 1\rangle = 1 + \beta = 0$, so $p_1 = x - 1$. For $p_2 = x^2 + \beta x + \gamma$, the two conditions $\langle p_2, 1\rangle = 0$ and $\langle p_2, x\rangle = 0$ read

$$2 + \beta + \gamma = 0, \qquad 6 + 2\beta + \gamma = 0.$$

Subtracting gives $\beta = -4$, then $\gamma = 2$. So $p_2 = x^2 - 4x + 2$, whose roots $2 \pm \sqrt{2}$ we already located in the Quick check above.

Exercise. B3: One more Legendre polynomial

Continue Example (The monic Legendre polynomials, continued): compute $\langle p_3, p_3\rangle$ for $p_3 = x^3 - \frac{3}{5}x$, then use Theorem 11.10.3 to find p_4. Verify that $\langle p_4, 1\rangle = \langle p_4, x^2\rangle = 0$.

Solution.

$$\langle p_3, p_3\rangle = \int_{-1}^{1} \Big(t^6 - \tfrac{6}{5}t^4 + \tfrac{9}{25}t^2\Big)\mathrm{d}t = \tfrac{2}{7} - \tfrac{6}{5}\cdot\tfrac{2}{5} + \tfrac{9}{25}\cdot\tfrac{2}{3} = \tfrac{2}{7} - \tfrac{6}{25} = \tfrac{8}{175}.$$

Since the weight is even, $a_3 = 0$, and $b_3 = \langle p_3, p_3\rangle / \langle p_2, p_2\rangle = (8/175)/(8/45) = \tfrac{9}{35}$. Hence

$$p_4 = x\Big(x^3 - \tfrac{3}{5}x\Big) - \tfrac{9}{35}\Big(x^2 - \tfrac{1}{3}\Big) = x^4 - \tfrac{6}{7}x^2 + \tfrac{3}{35}.$$

Verification. $\langle p_4, 1\rangle = \tfrac{2}{5} - \tfrac{6}{7}\cdot\tfrac{2}{3} + \tfrac{3}{35}\cdot 2 = \tfrac{14-20+6}{35} = 0$, and

$$\langle p_4, x^2\rangle = \tfrac{2}{7} - \tfrac{6}{7}\cdot\tfrac{2}{5} + \tfrac{3}{35}\cdot\tfrac{2}{3} = \tfrac{10-12+2}{35} = 0.$$

Orthogonality to x and x^3 is automatic by parity.

Exercise. B4: A two-point rule on the unit interval

Work on $(0,1)$ with $w = 1$. Exercise B2 of Section 2 (Exercise (B2: Gram–Schmidt on polynomials)) found the monic orthogonal polynomials 1, $x - \tfrac{1}{2}$, $x^2 - x + \tfrac{1}{6}$.

(a) Write down the two-point Gaussian rule for $\int_0^1 f$.

(b) Check it on $f = x^3$, and show that it fails on $f = x^4$.

Solution.

(a) The nodes are the roots of $x^2 - x + \tfrac{1}{6}$, namely

$$t_{1,2} = \tfrac{1}{2} \mp \tfrac{\sqrt{3}}{6}, \qquad t_2 - t_1 = \tfrac{\sqrt{3}}{3},$$

both in $(0,1)$, as Theorem 11.10.5 promises. With $\ell_1 = (x-t_2)/(t_1-t_2)$ and $\int_0^1 t\,\mathrm{d}t = \tfrac{1}{2}$,

$$\alpha_1 = \frac{\tfrac{1}{2} - t_2}{t_1 - t_2} = \frac{-\sqrt{3}/6}{-\sqrt{3}/3} = \tfrac{1}{2},$$

and $\alpha_2 = \tfrac{1}{2}$ likewise. So $\int_0^1 f = \tfrac{1}{2}(f(t_1) + f(t_2))$ for $f \in \mathbb{R}[x]_{\le 3}$.

(b) Write $t_{1,2} = \tfrac{1}{2} \mp u$ with $u = \sqrt{3}/6$, so $u^2 = \tfrac{1}{12}$. The odd powers of u cancel in a sum over t_1 and t_2, so

$$t_1^3 + t_2^3 = 2\Big(\tfrac{1}{8} + 3\cdot\tfrac{1}{2}\cdot\tfrac{1}{12}\Big) = 2\cdot\tfrac{1}{4} = \tfrac{1}{2},$$

and half of that is $\tfrac{1}{4} = \int_0^1 t^3\,\mathrm{d}t$. For the quartic,

$$t_1^4 + t_2^4 = 2\Big(\tfrac{1}{16} + 6\cdot\tfrac{1}{4}\cdot\tfrac{1}{12} + \tfrac{1}{144}\Big) = 2\cdot\tfrac{28}{144} = \tfrac{7}{18},$$

so the rule returns $\tfrac{7}{36}$, whereas $\int_0^1 t^4\,\mathrm{d}t = \tfrac{1}{5}$. The rule is exact to degree $3 = 2k-1$ and no further.

11.10.7.3 C. Going deeper

Exercise. C1: Interlacing roots

Let w be a weight on (a, b) and let (p_k) be its monic orthogonal polynomial sequence. The roots of p_k are said to **interlace** those of p_{k+1} when exactly one root of p_k lies strictly between each pair of consecutive roots of p_{k+1}. This is true for every k; prove the first two cases by hand.

(a) Let r be the root of p_1 and $s_1 < s_2$ the roots of p_2. Prove that $s_1 < r < s_2$.

(b) Let $u_1 < u_2 < u_3$ be the roots of p_3. Prove that $u_1 < s_1 < u_2 < s_2 < u_3$.

Hint: evaluate the three-term recurrence at a root.

Solution.

(a) By Theorem 11.10.3 with $k = 1$, $p_2 = (x-a_1)p_1 - b_1 p_0 = (x-a_1)(x-r) - b_1$. Evaluating at r gives $p_2(r) = -b_1 < 0$, since $b_1 > 0$. By Theorem 11.10.5, $p_2 = (x - s_1)(x - s_2)$ with $s_1 < s_2$ real, and this product is negative exactly on (s_1, s_2). Hence $s_1 < r < s_2$.

(b) By Theorem 11.10.3 with $k = 2$, $p_3 = (x - a_2)p_2 - b_2 p_1$. At a root of p_2 the first term dies, so

$$p_3(s_i) = -b_2\, p_1(s_i) = -b_2(s_i - r) \qquad (i = 1, 2).$$

By (a), $s_1 - r < 0$ and $s_2 - r > 0$, and $b_2 > 0$, so $p_3(s_1) > 0$ and $p_3(s_2) < 0$.

By Theorem 11.10.5, $p_3 = (x - u_1)(x - u_2)(x - u_3)$ with $u_1 < u_2 < u_3$. Such a product is negative on $(-\infty, u_1)$, positive on (u_1, u_2), negative on (u_2, u_3), and positive on (u_3, ∞). So $p_3(s_1) > 0$ puts s_1 in (u_1, u_2) or in (u_3, ∞), and $p_3(s_2) < 0$ puts s_2 in $(-\infty, u_1)$ or in (u_2, u_3). The option $s_1 > u_3$ is impossible: it would give $s_2 > s_1 > u_3$, where p_3 is positive, contradicting $p_3(s_2) < 0$. The option $s_2 < u_1$ is impossible too: it would give $s_1 < s_2 < u_1$, where p_3 is negative, contradicting $p_3(s_1) > 0$. Hence $u_1 < s_1 < u_2$ and $u_2 < s_2 < u_3$, which is the assertion.

The general case is not proved this way. The coefficients a_k and b_k assemble into a symmetric tridiagonal matrix whose characteristic polynomial is p_k, and the interlacing then follows from the interlacing theorem for the eigenvalues of a symmetric matrix and its leading submatrices, which is Chapter 17's business.

Exercise. C2: The ceiling is 2k − 1

Let w be a weight on (a, b) and let $k \geq 1$.

(a) Let $t_1, \dots, t_k$ be **any** k distinct real numbers and $\alpha_1, \dots, \alpha_k$ **any** real numbers. Prove that there is an $f \in \mathbb{R}[x]_{\leq 2k}$ with $\int_a^b f w \neq \sum_i \alpha_i f(t_i)$.

(b) Hence deduce that the Gaussian rule of Theorem 11.10.6 is exact on $\mathbb{R}[x]_{\leq 2k-1}$ and on no larger space $\mathbb{R}[x]_{\leq d}$.

Solution.

(a) Put $g = (x - t_1)\cdots(x - t_k)$ and $f = g^2$, a polynomial of degree $2k$. Every node is a root of g, so $f(t_i) = 0$ for every i and the right-hand side is 0, whatever the α_i are. The left-hand side is $\int_a^b g^2 w = \langle g, g\rangle$, and $g \neq 0$, so $\langle g, g\rangle > 0$ by positive definiteness.

The two sides differ.

(b) Exactness on $\mathbb{R}[x]_{\le 2k-1}$ is Theorem 11.10.6. Suppose the rule were exact on $\mathbb{R}[x]_{\le d}$ for some $d \ge 2k$. Since $\mathbb{R}[x]_{\le 2k} \subseteq \mathbb{R}[x]_{\le d}$, it would then be exact on every polynomial of degree at most $2k$. But its nodes are k distinct real numbers (Theorem 11.10.5), so (a) supplies a polynomial of degree $2k$ on which it is not. Hence no such d exists, and $2k-1$ is the largest degree for which the rule is exact. The concrete instance $k = 2$, $f = x^4$ is the failure computed in Example (Two-point Gauss–Legendre).